Structural Bioinformatics

ELIXIR CZECH REPUBLIC

Structural bioinformatics is an important contributor to understanding the molecular basis of biology and medicine, since structures define the function of each biological agent. Structural bioinformatics is traditionally strong in the Czech Republic due to historical links to the PDB database, top-class X-ray, NMR and Cryo-EM groups, and groups developing structural bioinformatics software tools and databases used worldwide. ELIXIR CZ initiated the formation of the ELIXIR 3D-Bioinfo community and is deeply involved in the activities of this community.

Key domains

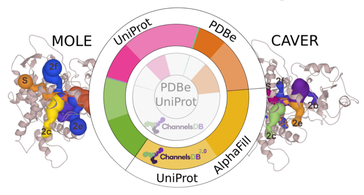
- The visualization of biomacromolecular structures
- Analysis and validations of biomacromolecular structures
- Protein engineering tools
- Nucleic acid structure research

Strategic goals

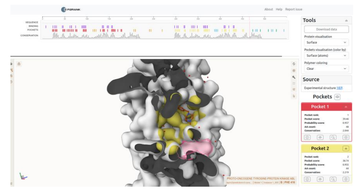
- Evolve and progress the key domains and services
- Work on activities and the growth of the ELIXIR 3D-Bioinfo community
- Synergy identification and ELIXIR CZ branding
- AI/ML approaches (AlphaFold)

Analysis of Biomolecular Structures

- Protein channels ChannelsDB 2.0
- Protein active sites P2RANK

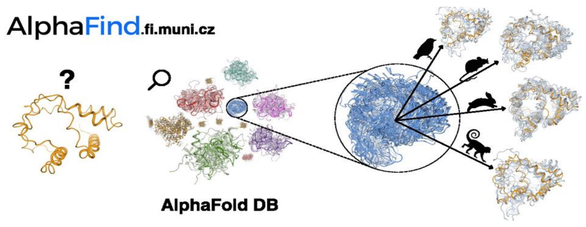


channelsdb2.biodata.ceitec.cz

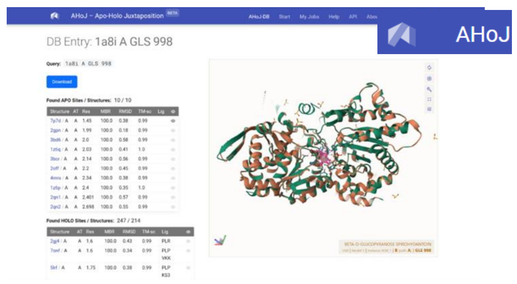


prankweb.cz

- Homologous protein models

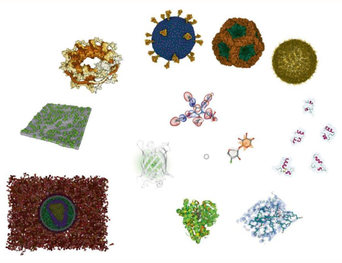


alphafind.fi.muni.cz

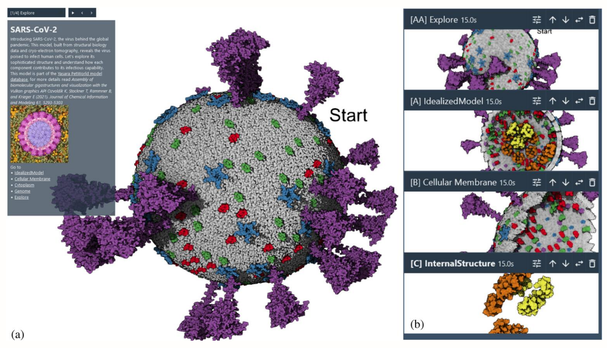


apoholo.cz

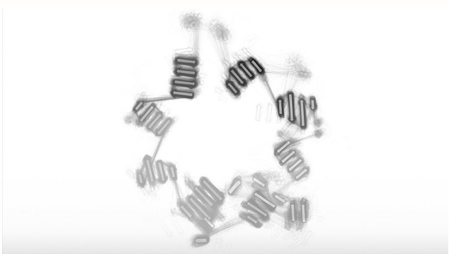
Visualization



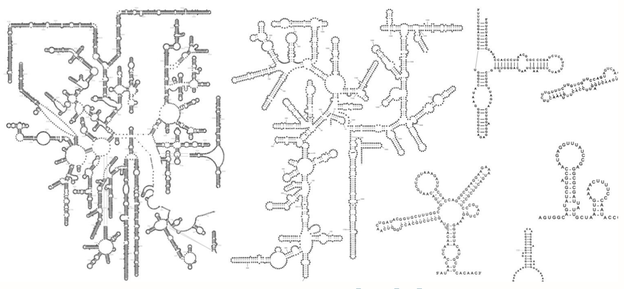
molstar.org
Quick web 3D visual



Mesoscale Explorer - molstar.org/me
interactive educational tours



2dprots.ncbr.muni.cz
2D protein families



r2dt.bio
R2DT nucleic acid 2D visualization

ELIXIR Communities

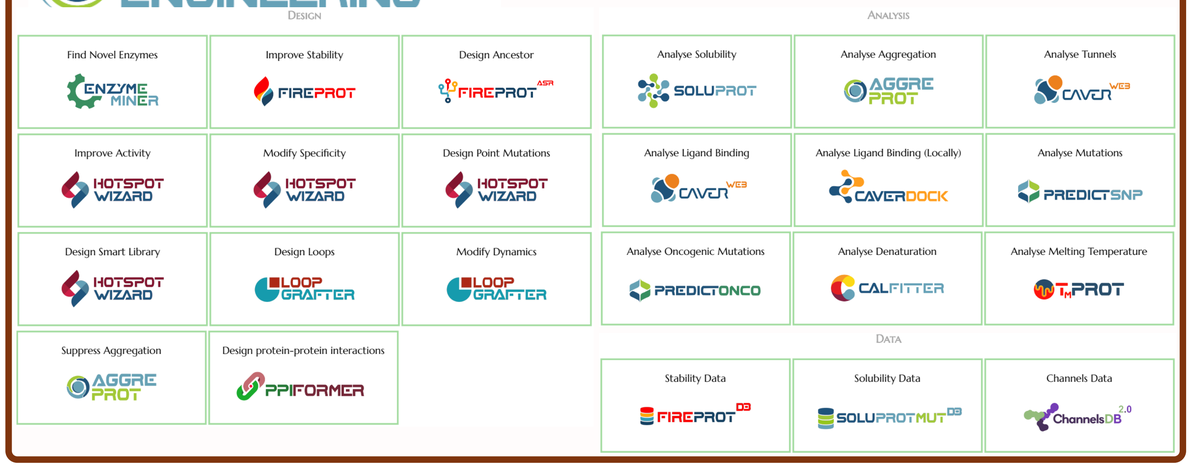
- **3Dbioinfo**
  Aims to study biomacromolecular 3D structural data
- **IDP**
  Aims to study intrinsically disordered proteins

Collaborations

- **ELIXIR Core resources - PDB** (Mol*)**, CATH** (2Dprots)**, PDBe-KB** (Mol*, ChannelsDB, P2Rank, FireProtDB)
- **Protein Channels** (MOLE + Caver + ChannelsDB)
- **Protein binding sites** (P2Rank + MOLE)

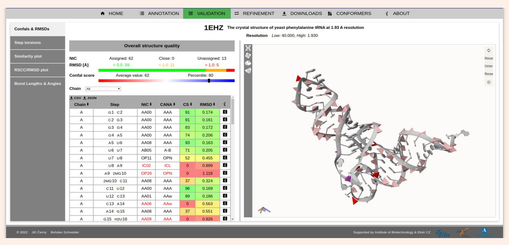
PROTEIN PORTAL ENGINEERING

loschmidt.chemi.muni.cz/portal

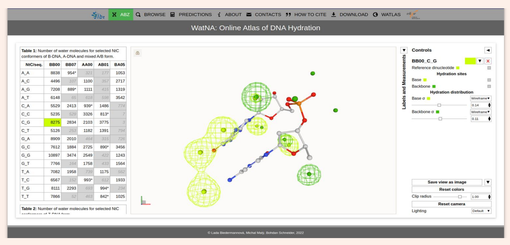


Nucleic Acids Structures

datmos.org



dnatco.datmos.org
DNA/RNA conformer analysis



watna.datmos.org
Water Atlas in DNA/RNA

Significant achievements

- Several **Nucleic Acid Res** papers (αCharges, ACC II, CATH with 2DProts, DNATCO, INTAA, LoopGrafter, FireProtDB, AggreProt, Mol*, Mol*VS, PrankWeb, "When will RNA get its AlphaFold moment?", AlphaFind, ChannelsDB 2.0)
- **Bioinformatics** papers (2DProts, AHoJ, CaverDock, OverProt, pyCaverDocs) and others
- Integrations to AlphaFold-DB, CATH, PDBe, PDBe-KB, RCSB PDB, RNAcentral, UniProt

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These projects have received funding from the European Union's Horizon 2020 research and innovation programme

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